COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE010	**SECTION:**	CPE21S2
DATE PERFORMED:	7/8/2026	**DATE SUBMITTED:**	7/15/2026
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ACTIVITY NO. HOA 1.1 Basic C++ Programming

1. PROCEDURE OUTPUT

What procedures did you perform?

ILO 1

HOA 1.1 > ILO1.cpp > main()

```cpp
#include<iostream>
int SumofTwoNum(int fNum, int sNum);
int isGreaterThan(float num1, float num2);
bool BooleanValues(bool A, bool B);

int SumofTwoNum(int fNum, int sNum) {
    return fNum + sNum;
}
int isGreaterThan(float num1, float num2){
    return num1 > num2;
}
bool BooleanValues(bool A, bool B){
    std::cout<<"A and B: "<< (A && B)<<std::endl;
    std::cout<<"A or B: "<< (A || B)<<std::endl;
    std::cout<<"Not A: "<< (!A) <<std::endl;
    std::cout<<"Not B: "<< (!B) <<std::endl;
    std::cout<<"A xor B: "<< (A ^ B)<<std::endl;

    return true;
}

int main() {
    std::cout<<"The of the Two Numbers is: "<<SumofTwoNum(6, 7)<<std::endl;
    std::cout<<"Is 1 Greater Than Two?: "<<isGreaterThan(1, 2)<<std::endl;
    if(BooleanValues(true, false)){
        std::cout<<"Displaying All Logical Operations"<<std::endl;

    }
    return 0;
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
The of the Two Numbers is: 13
Is 1 Greater Than Two?: 0
A and B: 0
A or B: 1
Not A: 0
Not B: 1
A xor B: 1
Displaying All Logical Operations
PS C:\Users\pagan\Downloads\HOA 1.1-20260714T134555Z-1-001\HOA 1.1>
```

Sections	Answer
Header File Declaration Section	HOA 1.1 > ILO1.cpp > main() `#include<iostream>`

Global Declaration Section	`int SumofTwoNum(int fNum, int sNum);`
	`int isGreaterThan(float num1, float num2);`
	`bool BooleanValues(bool A, bool B);`

Class Declaration and Method Definition Section	

Main Function	`int main() {`
	`std::cout<<"The of the Two Numbers is: "<<SumofTwoNum(6, 7)<<std::endl;`
	`std::cout<<"Is 1 Greater Than Two?: "<<isGreaterThan(1, 2)<<std::endl;`
	`if(BooleanValues(true, false)){`
	`std::cout<<"Displaying All Logical Operations"<<std::endl;`
	`}`
	`return 0;`
	`}`

Method Definition	`int SumofTwoNum(int fNum, int sNum) {`
	`return fNum + sNum;`
	`}`
	`int isGreaterThan(float num1, float num2){`
	`return num1 > num2;`
	`}`
	`bool BooleanValues(bool A, bool B){`
	`std::cout<<"A and B: "<< (A && B)<<std::endl;`
	`std::cout<<"A or B: "<< (A || B)<<std::endl;`
	`std::cout<<"Not A: "<< (!A) <<std::endl;`
	`std::cout<<"Not B: "<< (!B) <<std::endl;`
	`std::cout<<"A xor B: "<< (A ^ B)<<std::endl;`
	`return true;`
	`}`

ILO 2


HOA 1.1 > ILO2.cpp > setAngles(double, double, double)
#include<iostream>
class Triangle{
private:
double totalAngle, angleA, angleB, angleC;
double sideA, sideB, sideC;
public:
Triangle(double A, double B, double C);
void setAngles (double A, double B, double C);
const bool validateTriangle();
};
int main(){
Triangle setl(40, 30, 110);
if(setl.validateTriangle()){
std::cout<<"The shape is a valid triangle.\n";
}
else {
std::cout<<"The shape is NOT a valid triangle.\n";
}
return 0;
}
Triangle::Triangle(double A, double B, double C){
angleA = A;
angleB = B;
angleC = C;
totalAngle = A + B + C;
}
void Triangle::setAngles(double A, double B, double C){
angleA = A;
angleB = B;
angleC = C;
totalAngle = A + B + C;
}
const bool Triangle::validateTriangle(){
return (totalAngle <= 180);
}
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
PS C:\Users\pagan\Downloads\HOA 1.1-20260714T134555Z-1-001> cd "c:\Users\pagan\Downloads\HOA 1.1-
The shape is a valid triangle.
PS C:\Users\pagan\Downloads\HOA 1.1-20260714T134555Z-1-001\HOA 1.1>

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.
ILO 3 Part 1

HOA 1.1 > ILOC1.cpp > main()

```cpp
/*ILO 1*/
#include <iostream>

void SwapVar(int &var1, int &var2) {
    int temp = var1;
    var1 = var2;
    var2 = temp;
}

int main(){
    int a = 1;
    int b = 10;
    std::cout<<"Before Swapping A = "<<a<<" And B = "<<b<<"\n";

    SwapVar(a,b);
    std::cout<<"After Swapping A = "<<a<< " And B = " <<b;

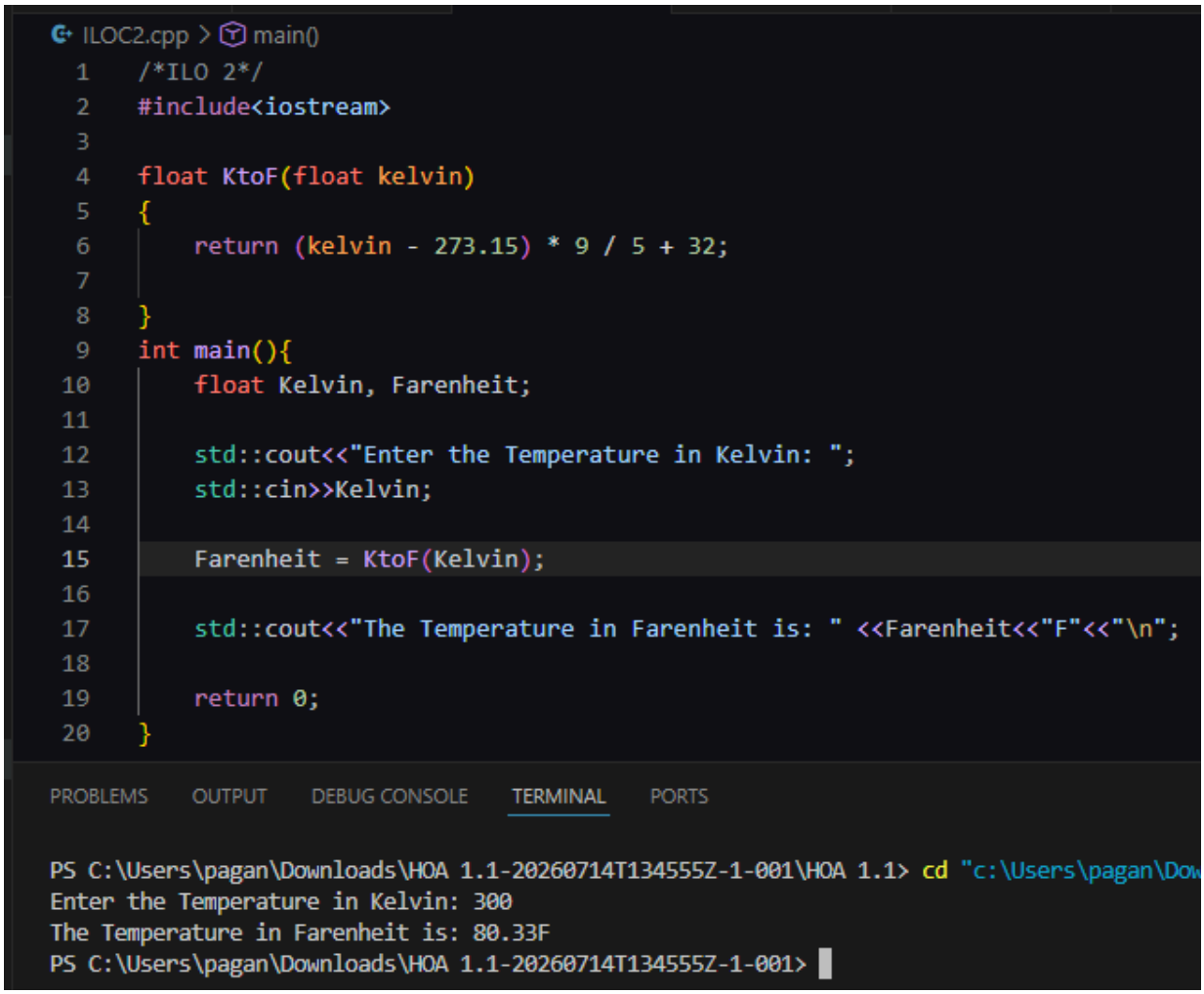
    return 0;
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\pagan\Downloads\HOA 1.1-20260714T134555Z-1-001\HOA 1.1> cd "c:\Users\pa
Before Swapping A = 1 And B = 10
After Swapping A = 10 And B = 1
PS C:\Users\pagan\Downloads\HOA 1.1-20260714T134555Z-1-001\HOA 1.1>
```

In this code, I used a void function in order to create temporary variables. After that, I used a temp command and assigned it in my variable 1. This temp command allows a string to hold a temporary variable and will be switched after the 1st variable is swapped. In my main command, I printed out first the “before swapping” after printing the after swapping. I called the function named “SwapVar” in order to execute the swapping of variables.

Part 2



In this code, I first declared a function in which it computes any given kelvin temperature to fahrenheit. In my main directory, I assigned “Kelvin and Farenheit” as a float, then asked for the temperature of the kelvin. After getting the given kelvin, I assigned the fahrenheit conversion to my function in order to get the final answer.

Part 3

ILOC3.cpp > main()

```cpp
/*ILO 3*/
#include<iostream>
#include<cmath>

float DistanceFormula(float x1, float x2, float y1, float y2){
    return sqrt(pow(x2 - x1, 2)+pow(y2 - y1, 2));
}

int main(){
    float x1, x2, y1, y2, Distance;
    std::cout<<"Enter Value for X1: ";
    std::cin>> x1;
    std::cout<<"Enter Value for X2: ";
    std::cin>> x2;
    std::cout<<"Enter Value for Y1: ";
    std::cin>> y1;
    std::cout<<"Enter Value for Y2: ";
    std::cin>> y2;

    Distance = DistanceFormula(x1, x2, y1, y2);

    std::cout<<"The Distance Between Two points is: "<<Distance<<"\n";

    return 0;
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\pagan\Downloads\HOA 1.1-20260714T134555Z-1-001> cd "c:\Users\pagan\Downlo
Enter Value for X1: 5
Enter Value for X2: 10
Enter Value for Y1: 15
Enter Value for Y2: 20
The Distance Between Two points is: 7.07107
PS C:\Users\pagan\Downloads\HOA 1.1-20260714T134555Z-1-001>
```

I used a function “DistanceFormula” with a datatype of float; then I returned the formula of the distance formula. In my main directory, I asked for the value of x1, x2, y1, and y2; after that I assigned a new variable called “Distance” to be called after answering the problem.

Part 4

ILOC4.cpp > main()

```cpp
/*ILO 4*/
#include<iostream>
#include<cmath>
class Triangle {
    private:
        double totalAngle, angleA, angleB, angleC;
        double sideA, sideB, sideC;
    public:
        Triangle(double A, double B, double C, double a, double b, double c);
        void setAngles(double A, double B, double C);
        bool validateTriangle();
        double computeArea();
        double computePerimeter();
        void validateType();

};

Triangle::Triangle(double A, double B, double C, double a, double b, double c) {
    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = A+B+C;

    sideA = a;
    sideB = b;
    sideC = c;
}
void Triangle::setAngles(double A, double B, double C) {
    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = A+B+C;
}
bool Triangle::validateTriangle() {
    return (totalAngle <= 180);
}

double Triangle::computeArea() {
    double side = (sideA + sideB + sideC) / 2;
    return sqrt(side * (side - sideA)*(side - sideB)*(side - sideC));
}
double Triangle::computePerimeter() {
    return sideA + sideB + sideC;
}
void Triangle::validateType() {
    if (angleA < 90 && angleB < 90 && angleC < 90) {
        std::cout<<"The Angle is an Acute Angle"<<std::endl;
    }
    else if (angleA == 90 || angleB == 90 || angleC == 90) {
        std::cout<<"The Angle is a Right Angle"<<std::endl;
```

```
51
52      }
53      else if (angleA > 90 || angleB > 90 || angleC == 90) {
54          std::cout<<"The Angle is an Obtuse Angle"<<std::endl;
55      }
56      else {
57          std::cout<<"Other Angle of Not an Angel"<<std::endl;
58      }
59  }
60
61  int main()
62  {
63      Triangle setl(80, 80, 20, 25, 25, 25);
64      if (setl.validateTriangle()) {
65          std::cout<<"The Shape is a Triangle"<<std::endl;
66          std::cout<<"The Area is: "<<setl.computeArea()<<std::endl;
67          std::cout<<"The Perimeter is: "<<setl.computePerimeter()<<std::endl;
68
69          setl.validateType();
70
71      } else {
72          std::cout<<"The Shape is NOT a Triangle"<<std::endl;
73      }
74      return 0;
75  }
```

```
                                                    > cd
The Shape is a Triangle
The Area is: 270.633
The Perimeter is: 75
The Angle is an Acute Angle
PS C:\Users\pagan\Downloads\HOA 1.1-20260714T134555Z-1-001>
```

I added a function to solve for the area of the triangle given the 3 sides. I used a double data type and returned the formula for the area of the triangle. I also declare another function that solves its parameter. Lastly, I added a validateType that verifies if the triangle is an acute, right, or an obtuse angle. In the main function I called all the new functions to get the answers needed.

Conclusion:
I still needed more time to be comfortable in this language again. I always keep forgetting semicolons and the names of functions I just declared. But, I recalled the very basics of C++ like the if statements, functions, and classes. The supplementary activity is the hardest for me, I always forget there are available classes that can be used to solve math problems. Overall, I wanted to expand my knowledge in C++ more because I had absorbed little when I was in my 1st year.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

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I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **ChatGPT**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **I used chatgpt to search the distance formula and heron's formula. I also used AI on how to see the problem and fix it in the most minimal help of AI**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **I used AI to check if there are errors and did not ask it to fix it for me. I also used AI to gain specific knowledge about the 3rd part of the ILO 1 since I completely do not understand how it will operate, so I used AI to briefly explain it to me.**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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